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Registration No:	23-NTU-CS-1085
Course Name:	Embedded System and Iot
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ASSIGNMENT 01:

Question # 03:

TASK A:

Circuit Diagram: Design a Wokwi circuit and draw a neat hand-sketch including:

- 2 push buttons
- 3 LEDs
- 1 buzzer
- 1 OLED

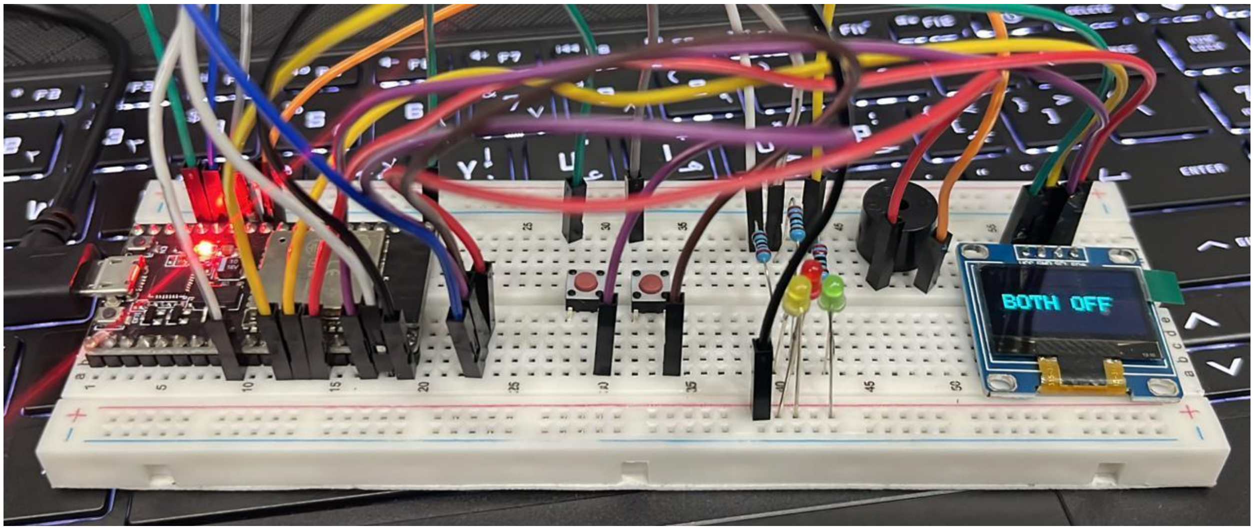
Task A — Coding: Use one button to cycle through LED modes (display the current state on

the OLED):

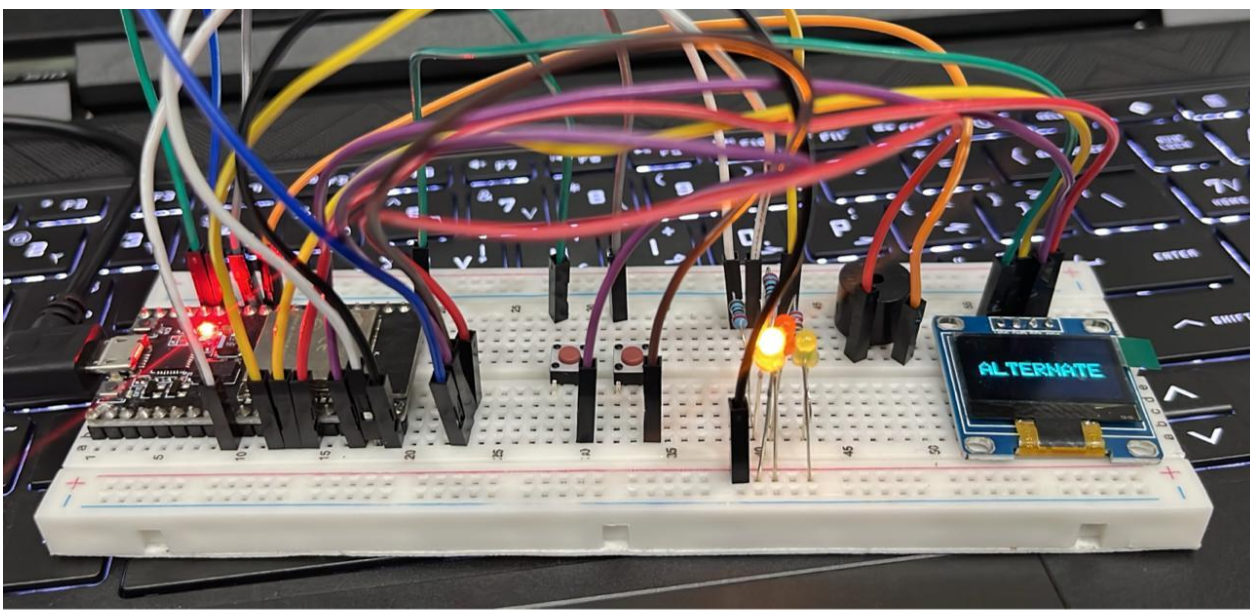
1. Both OFF
2. Alternate blink
3. Both ON
4. PWM fade

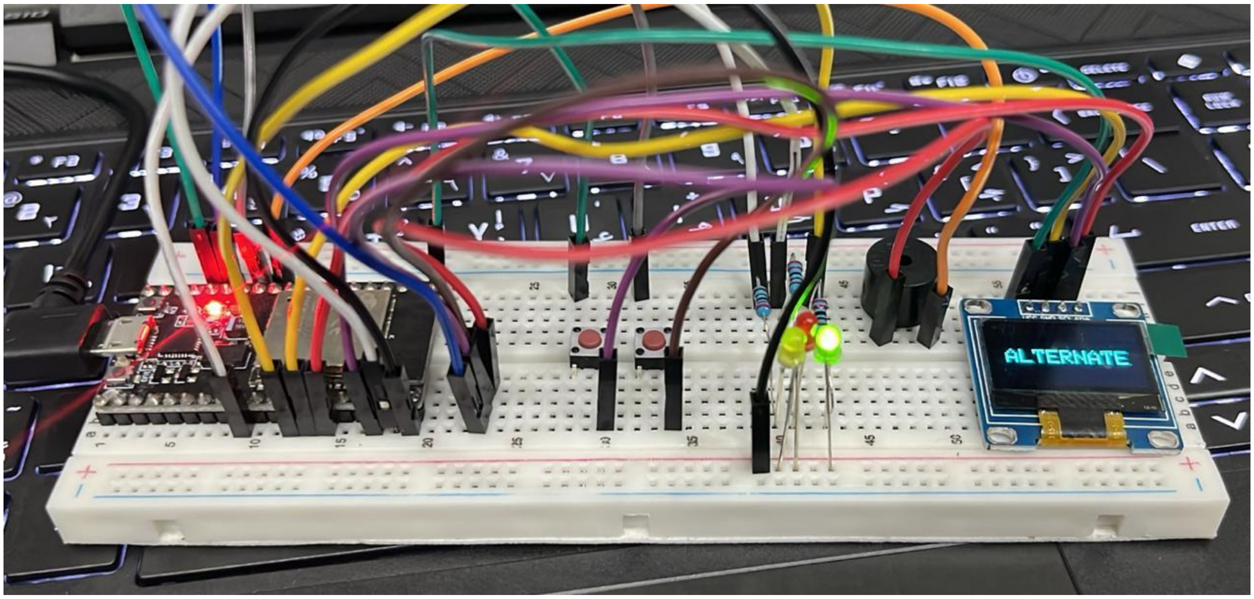
Use the second button to reset to OFF.

BOTH OFF:

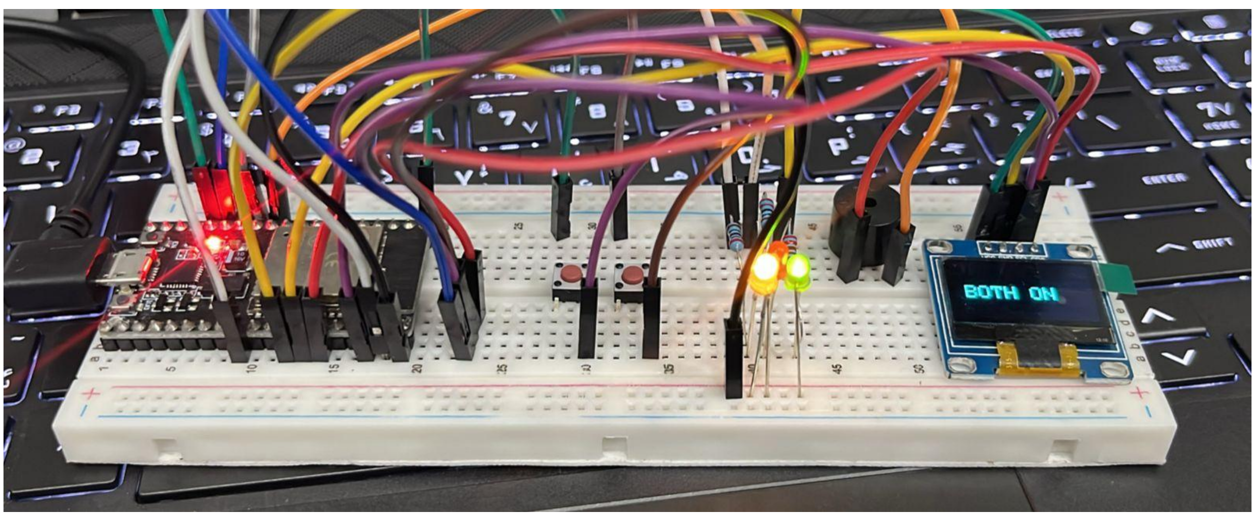


ALTERNATE:

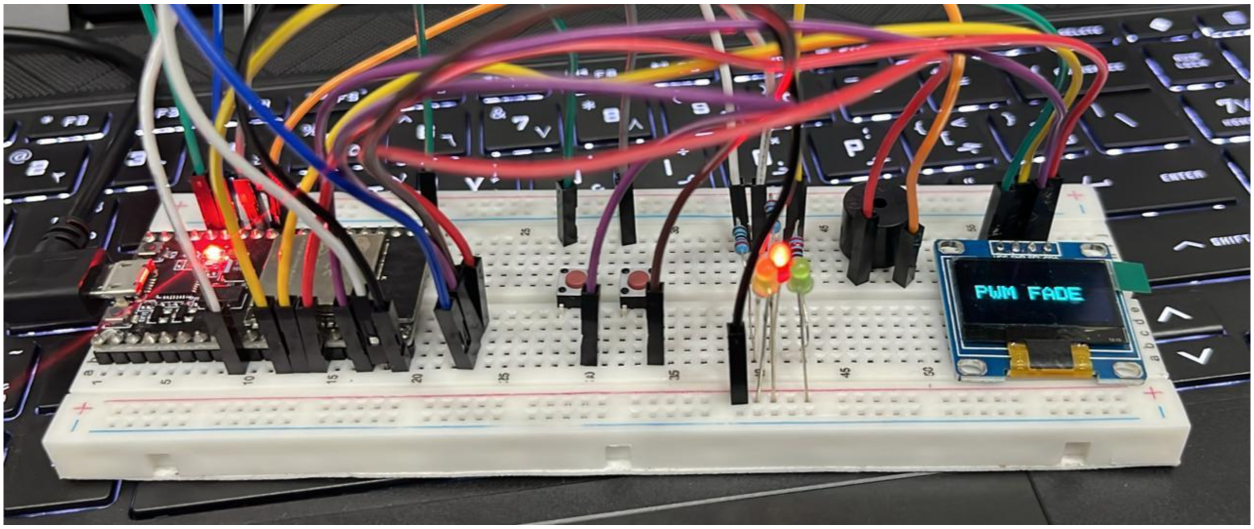




BOTH ON:



PWM FADE:



HANDWRITTEN CODE:

TASK A :-

```
#include<Arduino.h>
#include<Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#define LED1 23
# define LED2 19
# define LED3 17
# define BUTTON1 14
# define BUTTON2 13
# define DEBOUNCE_MS 50
# define DEBOUNCE_US (DEBOUNCE_MS*1000UL)
# define SCREEN_WIDTH 128
# define SCREEN_HEIGHT 64
# define OLED_ADDR 0x3C
hw_timer_t * debounceTimer = nullptr;
volatile bool debounceActive = false;
volatile int mode count = 0;
Adafruit_SSD 1306 display(SCREEN_WIDTH, SCREEN_HEIGHT,
&Wire, -1);
// Debounce timer ISR
void IRAM_ATTR onDebounceTimer () {
  if (digitalRead (BUTTON1) == LOW) {
    modeCount++;
    if (mode count > 3) mode count = 0;
  }
```

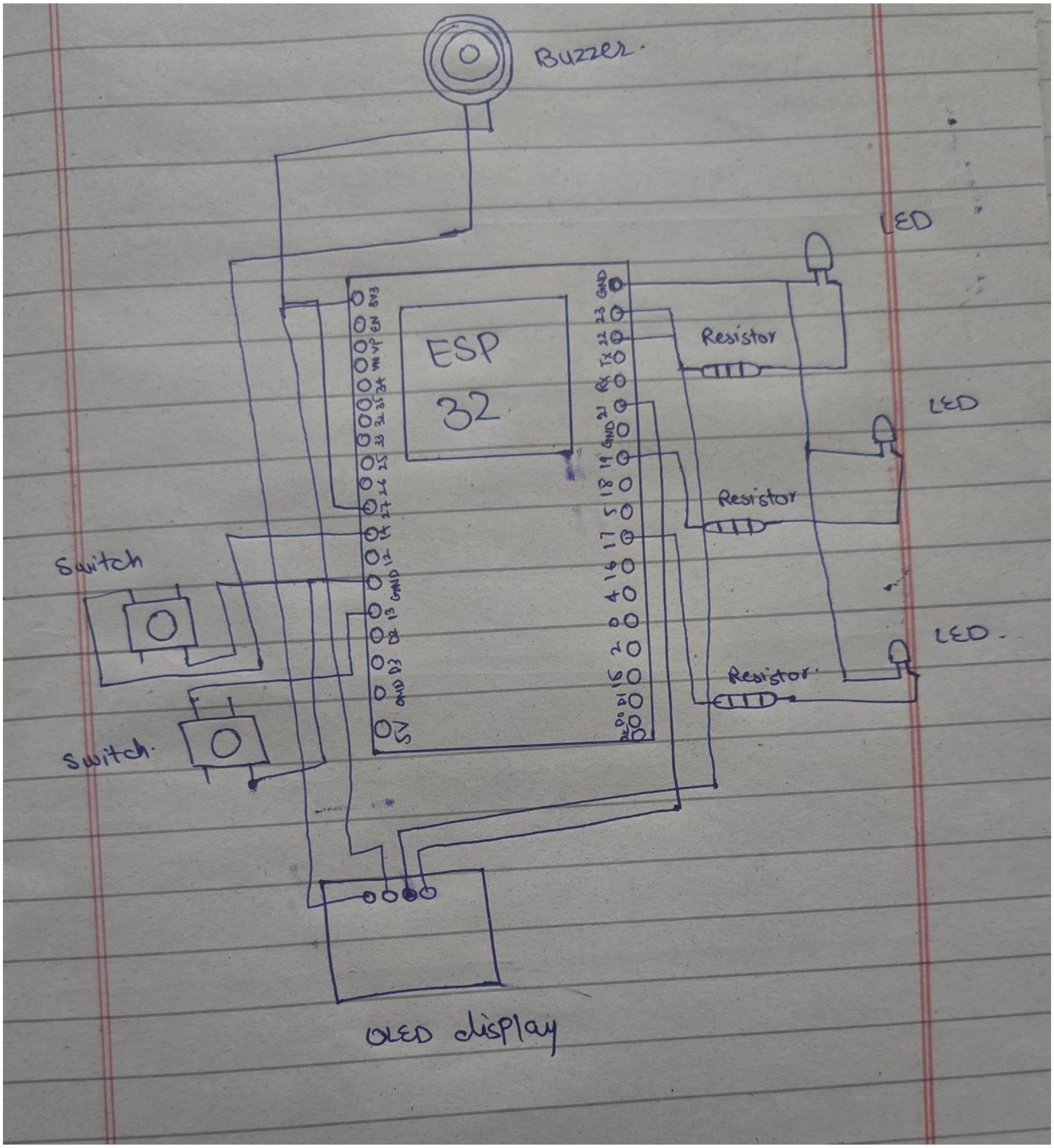
```
if (digitalRead (BUTTON2) ==Low)){
  modeCount = 0;
 }
 debounceActive = false;
}.

# // Button1 interrupt ISR.
void IRAM_ATTR onButton1SR(){
 if(! debounceActive){.
  debounceActive = true;
  timerWrite(debounceTimer, 0);
  timerAlarmWrite(debounceTimer, DEBOUNCE_US,
  false);
  timerAlarmEnable(debounceTimer);
 }
}
// Display helper
void showMode( const char* text){.
   display.clearDisplay();
   display.setTextSize(2);
   display.setTextColor( SSD1306_WHITE);
   display.setCursor( 10,26);
   display.println(text);
   display.display();
}
```

```
void setup() {
// PWM setup
ledcSetup(2, 5000, 8);
ledcAttachPin(LED3, 2);
pinMode(LED1, OUTPUT);
pinMode(LED2, OUTPUT);
digitalWrite(LED1, LOW);
digitalWrite(LED2, LOW);
pinMode(BUTTON1, INPUT_PULLUP);
pinMode(BUTTON2, INPUT_PULLUP);
attachInterrupt(BUTTON1, onButtonISR, FALLING);
attachInterrupt(BUTTON2, onButtonISR, FALLING);
debounceTimer = timerBegin(3, 80, true)
timerAttachInterrupt(debounceTimer, &onDebounceTimer, true);

// OLED setup
wire.begin(21, 22);
if (!display.begin(SSD1306_SWITCHCAPVCC, OLED_ADDR)) {
    for (;;);
}
display.clearDisplay();
showMode("All OFF");
}
```

```
void loop(){.
switch (modeCount){.
case 0: // All off
digitalWrite(LED1, LOW);
digitalWrite(LED2, LOW);
ledcWrite(2,0);
showMode(" BOTH OFF");
break;

case 1: //Alternative blink
showMode("Alternate");
digitalWrite(LED1, LOW);
digitalWrite(LED2, HIGH);
ledcWrite(2,0);
delay(400);
digitalWrite(LED1, HIGH);
digitalWrite(LED2, LOW);
ledcWrite(2,0);
delay(400);
break;
case 2: // All ON.
digitalWrite(LED1, HIGH);
digitalWrite(LED2, HIGH);
ledcWrite(2,0);
showmode("Both ON");
break;
```

```
case 3 : // PWM FADE.
  digitalwrite (LED1 , LOW);
  digitalwrite ( LED 2 , LOW);
   showMode ("PWM FADE");
 for (int d=0;  d <= 255 ;  d=d+5){
   ledcWrite (2, d);
   delay (10);
 }
  for (int d= 255 ; d >=0 ; d=d-5)
 {
   ledcWrite (2, d);
   delay 10);
 }
   break;
}
  delay (50);
}
```

LABELLED DIAGRAM:



WOKWI LINK:

TASK A:

https://wokwi.com/projects/445781067971415041

TASK B:

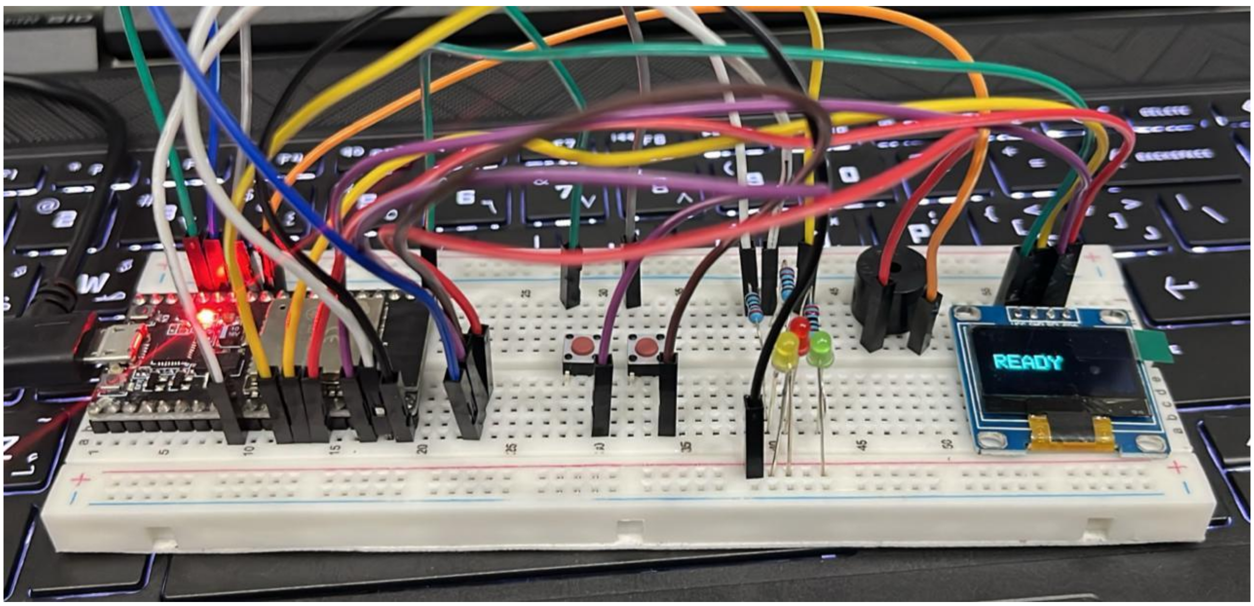
https://wokwi.com/projects/445790227147737089

Task B — Coding:

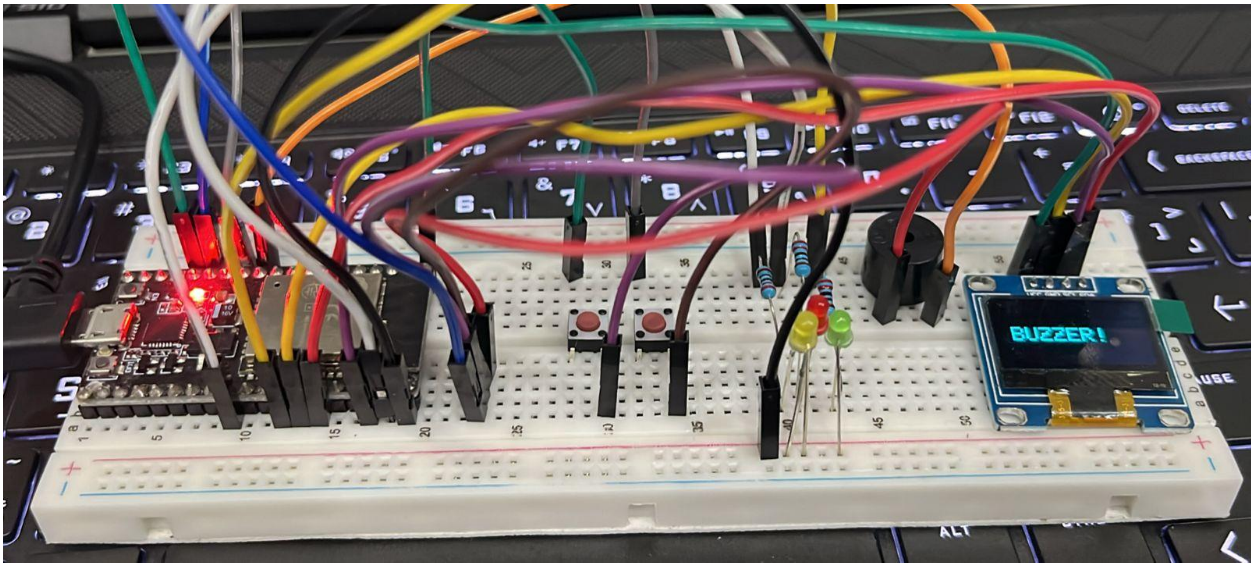
Use a single button with press-type detection (display the event on the OLED):

- **Short press → toggle LED**
- **Long press (> 1.5 s) → play a buzzer tone**

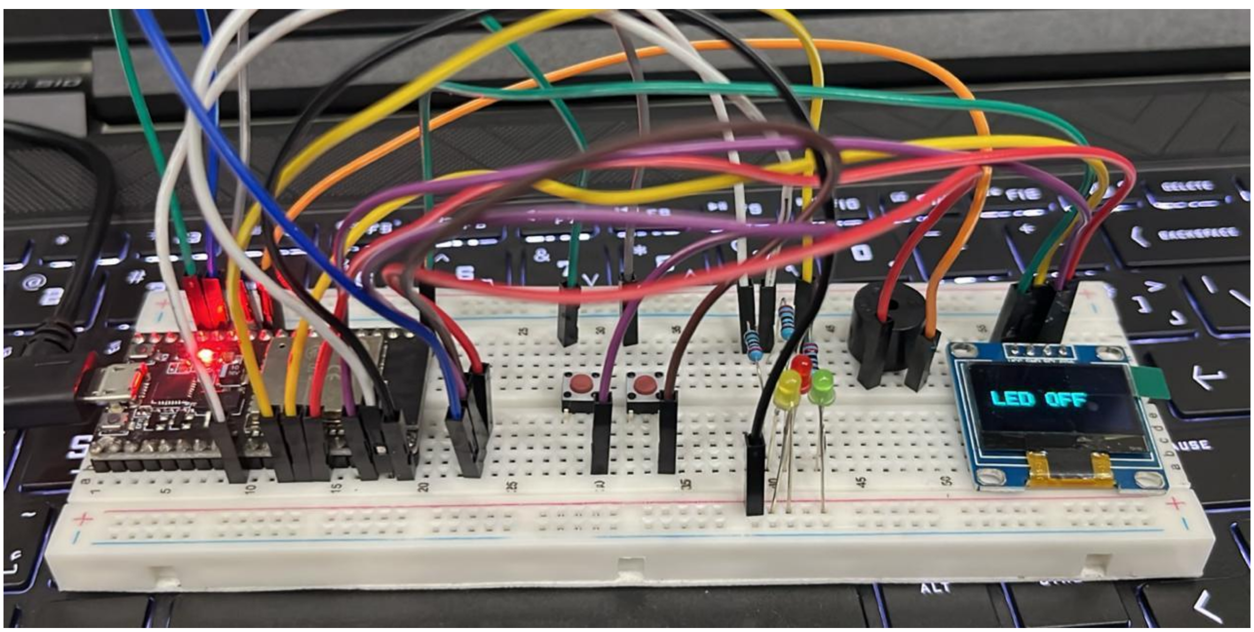
READY:



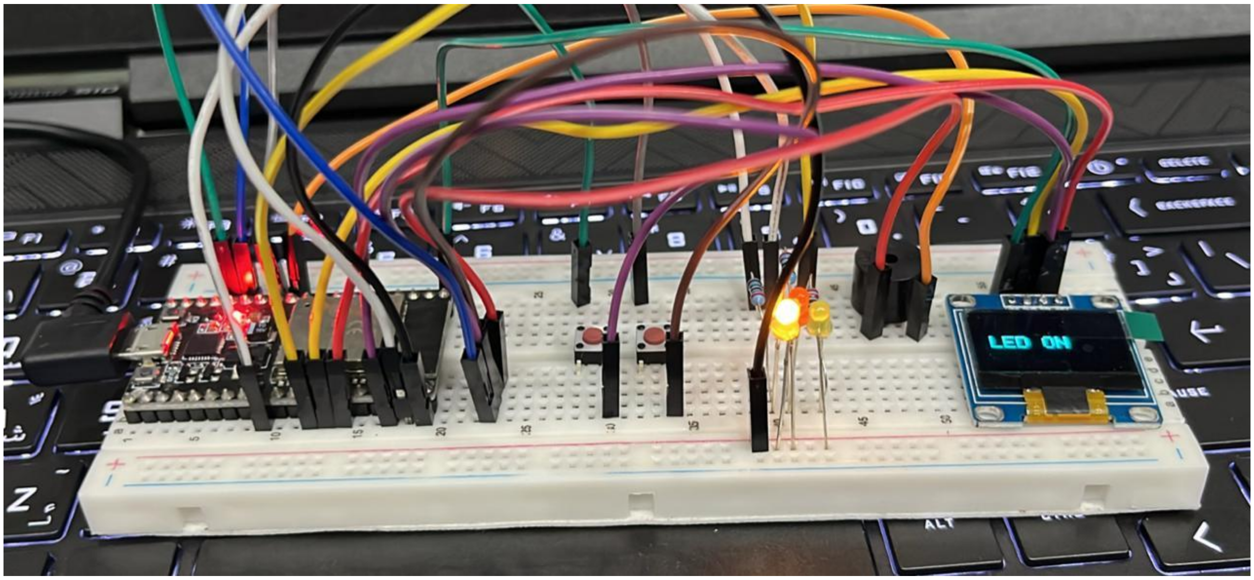
BUZZER:



LED OFF:



LED ON:



HANDWRITTEN CODE:

TASK B:-

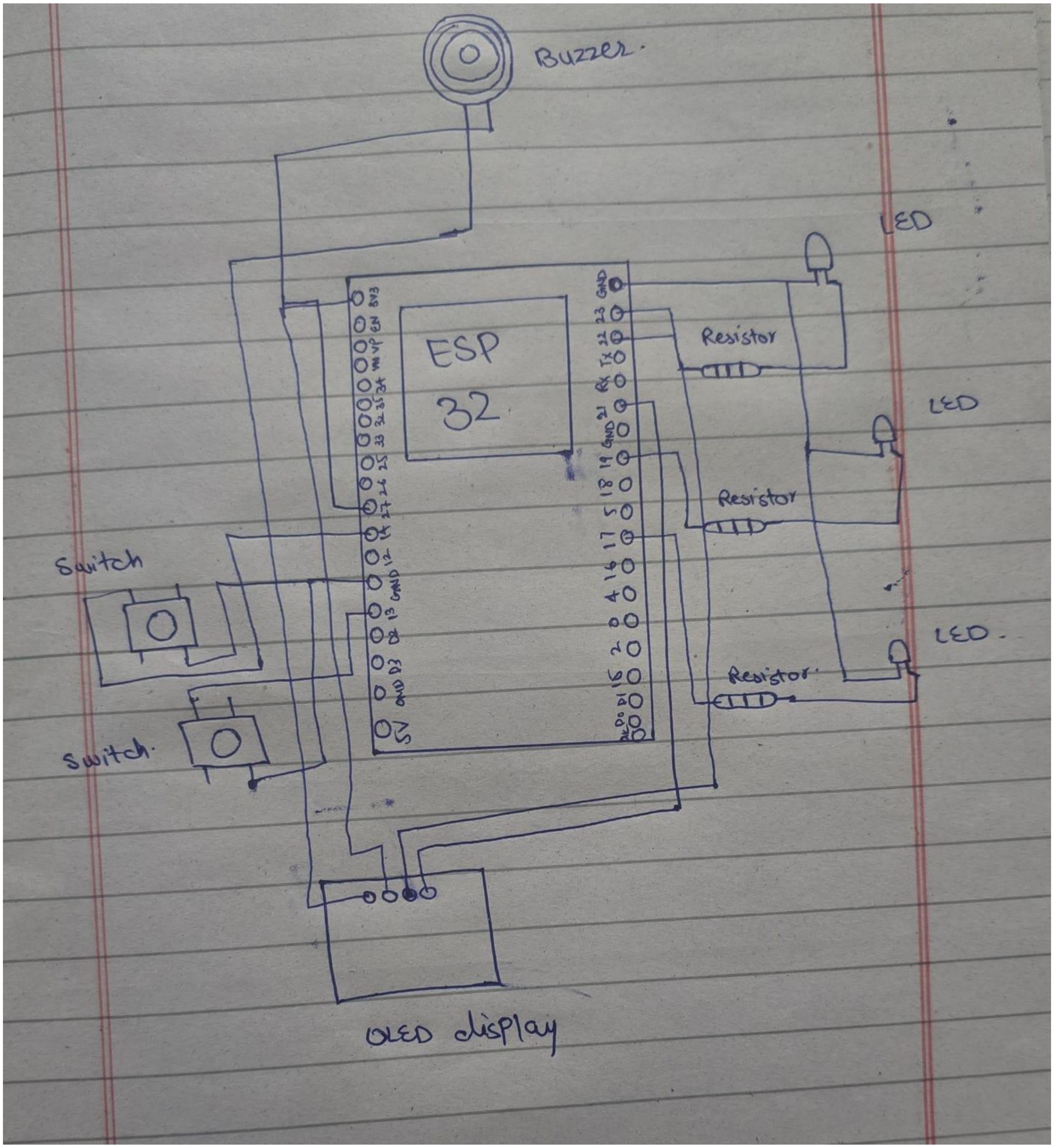
```
#include <Arduino.h>
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
// Pin configuration
#define LED1 23
#define Button1 14
#define BUZZER_PIN 27
// Buzzer configuration
#define PWM_CH 0
#define FREQ 2000
#define RESOLUTION 10
// OLED Config
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_ADDR 0x3C
Adafruit_SSD1306 display(SCREEN_WIDTH,
SCREEN_HEIGHT, &Wire, -1);
// Variables
bool ledstate = false;
unsigned long pressStart = 0;
bool buttonPressed = false;
void playBuzzerTone() {
  int melody[] = {
    330, 392, 330, 440, 494, 523, 392,
    262, 330, 392, 262, 196, 262, 330
  };
```

```
for ( int i = 0 ; i < 8 ; i++ ) {
    ledcWriteTone ( PWM_CH , melody [i] ) ;
    delay ( 200 ) ;
  }
  ledcWrite ( PWM_CH , 0 ) ;
}
void setup ( ) {
  pinMode ( LED1 , OUTPUT ) ;
  pinMode ( BUTTON1 , INPUT_PULLUP ) ;
// BUZZER SETUP
  ledcSetup ( PWM_CH , FREQ , RESOLUTION ) ;
  ledcAttachPin ( BUZZER_PIN , PWM_CH ) ;
// OLED SETUP
  Wire.begin ( 21 , 22 ) ;
  if ( ! display . begin ( SSD1306_SWITCHCAPVCC ,
OLED_ADDR ) ) {
      for ( ; ; ) ;
  }
  display . clearDisplay ( ) ;
  display . setTextSize ( 2 ) ;
  display . setTextColor ( SSD1306_WHITE ) ;
  display . setCursor ( 10 , 26 ) ;
  display . println ( "Ready" ) ;
  display . display ( ) ;
}
```

```
void loop(){
if(digitalRead(BUTTON1) ==LOW){.
  if(!ButtonPressed){.
    buttonPressed = true;.
    pressStart = millis();

  }
} else if (buttonPresed){.
  unsigned long pressDuration= millis()-
pressStart;.
  buttonpressed =false;.
if (pressDuration < 1500)
{   // short press: toggle LED
   ledstate = !ledstate;
  digitalWrite(LED1, ledstate ? HIGH :LOW;
  display.clearDisplay();
  display.setcursor(10,26);
   display.Println(ledState ? "LEDON":
"LED OFF");.
  display.display();
}
else{.
 // long press: play buzzer
 display.cleardisplay(),
 display.setcursor(10,26);
 display.println("BUZZER!");.
  display.display();
```

```
PlayBuzzerTone();
}
}.
delay(50);
}.
```

DIAGRAM:



GITHUB REPOSITORY LINK:

https://github.com/Qurat123567/Embedded---IOT-Systems.git